REFINEMENT vs. REFRESH

Compute allocation for closed-loop robot policies

Alba María Téllez Fernández

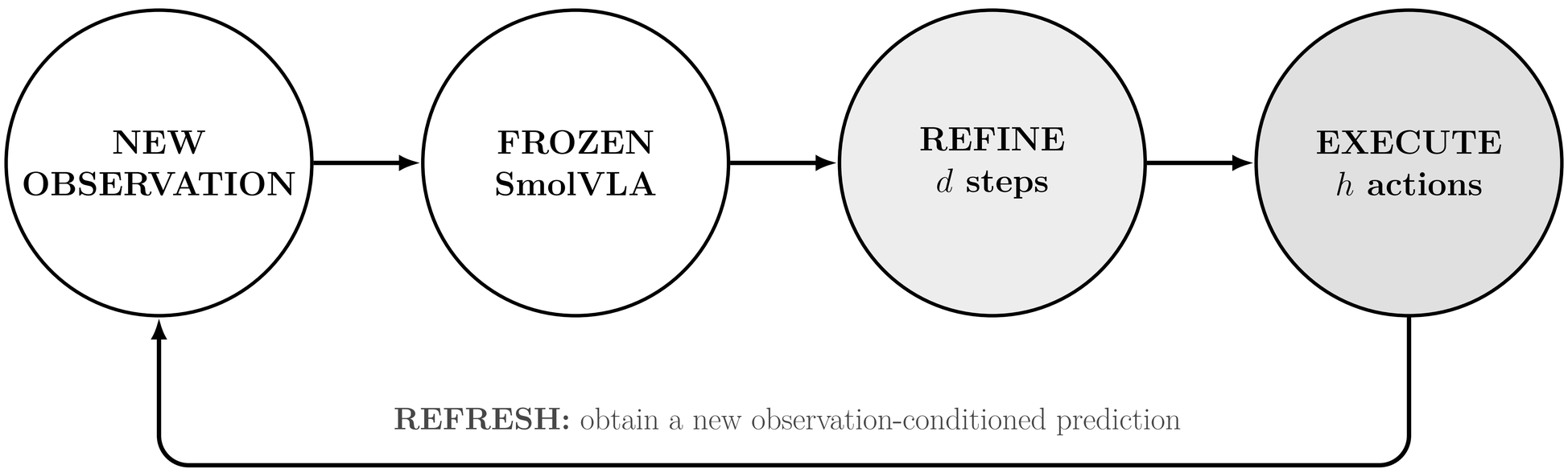
MAAT Institute • Zürich, 10 Nov 2026

INFERENCE COMPUTE IS A CONTROL DECISION.

THE SAME BUDGET HAS TWO JOBS

Refine the current action chunk—or generate again from a fresher observation.

WHAT SHOULD THE POLICY DO WITH ITS NEXT UNIT OF COMPUTE?



THE HELD-OUT RESULT

FEWER DENOISING STEPS. MORE SUCCESS.

Same two-action execution prefix; eight Goal and Object tasks excluded from configuration selection.

2 denoising steps

139 / 160 successes

5 denoising steps

122 / 160 successes

+10.6 pp

descriptive 95% interval
+4.4 to +16.9 pp

−5.9%
arithmetic per control step with 2 instead of 5 denoising steps

82.1%
of generation arithmetic is independent of denoising count

−60.7%
arithmetic per attempt for $h = 5$ versus $h = 2$, at $d = 2$

HOW WE TESTED IT

12 configurations
4 Spatial tasks
480 discovery attempts

7 frozen configurations
8 held-out tasks
1,120 confirmation attempts

One frozen checkpoint
Synchronous LIBERO
Paired evaluation

THE IDEA TO REMEMBER

Better robot inference is not only about refining longer. It is also about deciding when to look again.

Evaluate refinement and replanning jointly through closed-loop success and compute.

INTERPRET WITH CARE

Similar observed success for $h = 2$ and $h = 5$ does not establish equivalent quality. Results cover one checkpoint and synchronous simulation.

github.com/amtellezfernandez/refinement-refresh-robot-policies